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Empirical Testing of H-O Theory

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FPE AND THE DISTRIBUTION OF INCOME: A Case

- **For example – U.S. and India**
 - Trade opening up causes prices of machines and the prices of cloth to equalize between countries (H-O)
 - Size of machine and cloth industries will change for each country changing their ***industrial structure***



FACTOR-PRICE EQUALIZATION AND THE DISTRIBUTION OF INCOME

- *(Assume)* U.S. has a comparative advantage in machines ... (*machines are K intensive*)
 - This causes an increased demand for machines
 - The price of machines rises relative to price of cloth
 - Machine production expands
 - Cloth production contracts
 - Increased demand for inputs to make machines



FACTOR-PRICE EQUALIZATION AND THE DISTRIBUTION OF INCOME ... *(continued)*

- Increase in **capital** greater than increase in **labor** as machines are capital intensive
- Resources shift from cloth to machines
- Cloth industry declines
 - Imports replace much domestic production
 - More labor than capital released on market
 - Shortage of capital increase “profit” (*return to K*)
 - Surplus of labor decreases wages
- Ratio of wages to “profit” (*return to K*) declines



FACTOR-PRICE EQUALIZATION AND THE DISTRIBUTION OF INCOME

- India has a comparative advantage in cloth
 - Increased demand for cloth
 - Price of cloth rises relative to price of machines
 - Machine production contracts
 - Cloth production expands
 - Increased demand for inputs to make cloth



FACTOR-PRICE EQUALIZATION AND THE DISTRIBUTION OF INCOME

- Increase in labor greater than increase in capital as cloth is labor intensive
- Resources shift from machines to cloth
- Machine industry declines
 - Imports replace much domestic production
 - More capital than labor released on market
 - Shortage of labor increase wages
 - Surplus of capital decreases “profit”
- Ratio of wages to “profit” increases



FACTOR-PRICE EQUALIZATION AND THE DISTRIBUTION OF INCOME

- Wages
 - Decline in U.S.
 - Increase in India
- “Profits” (*return to capital*)
 - Increase in U.S.
 - Decrease in India

Overall – Factor Prices get closer to equalization (just as we saw with “Goods Prices” in H-O)



The Leontief Paradox

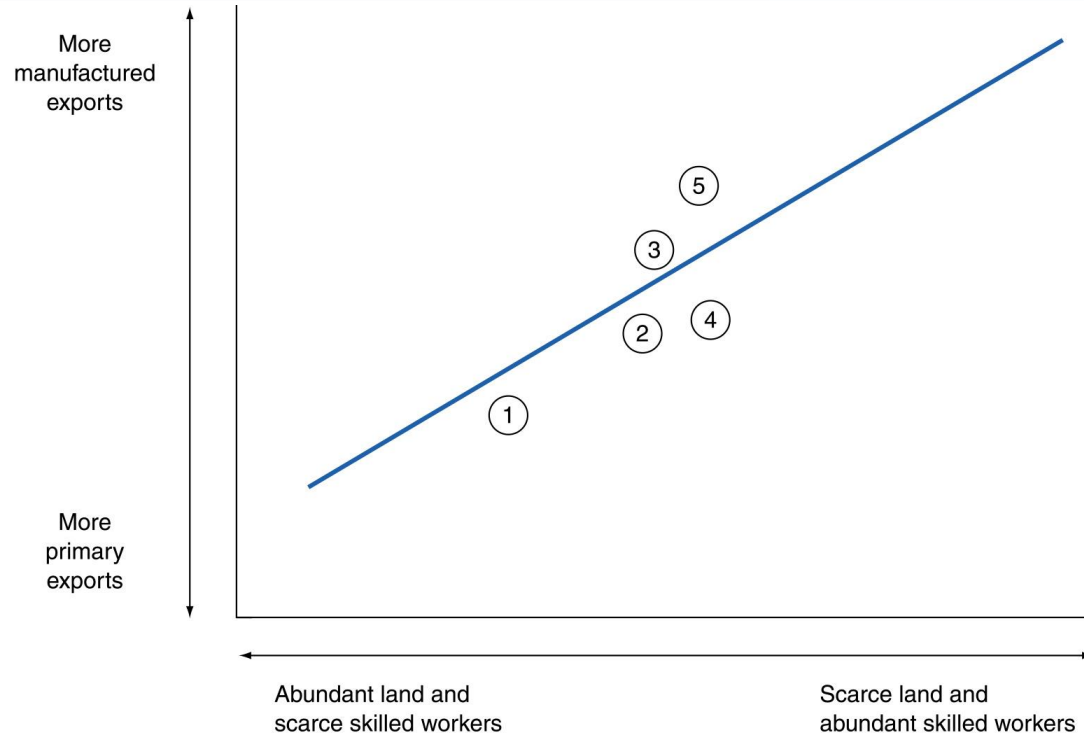
- A 1951 test of the H-O theory using US data for the year 1947 using **input-output table**
- Showed that the pattern of trade did not fit the conclusions of the H-O theorem.
 - Exports in the U.S. seemed to be labor intensive when they should have been capital intensive.



Source of the Leontief Paradox Bias

- Assumed a two factor world which required assumptions about what is capital and what is labor.
- Most heavily protected industries in U.S. were L-intensive, reduced imports and increased domestic production of L-intensive goods.
- Only physical capital included as capital, ignoring **human capital** (education, job training, skills).





Legend:

- (1) Sub-Saharan Africa; (2) Latin America and the Caribbean; (3) South Asia;
(4) East Asia and the Pacific; (5) Industrial market economies

FIGURE Comparative Advantage with Skills and Land.

Factor – Intensity Reversal (FIR)

- A given commodity is the labor-intensive commodity in the L-abundant nation and capital-intensive commodity in the K-abundant nation.
- The concept of Elasticity of Substitution (ES) is useful to measure this.
- Greater is the FIR, greater is the ES.
- FIR is neither present in H-O model nor also in factor equalization model (HOS, SS model)
- First empirical study was made by **Minhas in 1962**, found it is prevalent. Then studied by Leontief and Ball in 1964 and 1966 respectively



Qualification #1:

Factor Intensity Reversal

- Implicitly, we've assumed that if good X is K-intensive relative to good Y at one factor price ratio, it will be K-intensive at all factor prices
- A FIR is when a good is relatively K-intensive at one set of factor prices, but relatively labor intensive at another



Qualification #2: Factor Intensity Reversal

- FIRs occur when capital and labor can be substituted more easily in the production of one good than another
- Example:
 - Suppose apple production is much more flexible
 - If w/r is very high, apple producers will choose a relatively K-intensive method (as compared with butane makers)
 - If w/r starts to fall, apple producers will start substituting L for K
 - Eventually, it's possible that apples might be produced in a relatively L-intensive manner (as compared with butane)



Factor Intensity Reversal: Implications for Trade

- Suppose France is K-abundant relative to Germany (that is $(K/L)^F > (K/L)^G$)
- This means that $(w/r)^F > (w/r)^G$
- Suppose further that there is a FIR: in France, at $(w/r)^F$ apples are produced relatively K-intensively but in Germany at $(w/r)^G$ apples are produced in a relatively L-intensive way



Factor Intensity Reversal: Implications for Trade

- If trade begins, according to the H-O theorem the relatively K-abundant country (France) will export the rel. K-intensive good (apples) and the rel. L-abundant country will export the rel. L-intensive good (also apples)
- This can't work



Factor Intensity Reversal: The Bottom Line

- If FIRs are present, the H-O theorem breaks down
- If FIRs are common, we'd better find a new theory

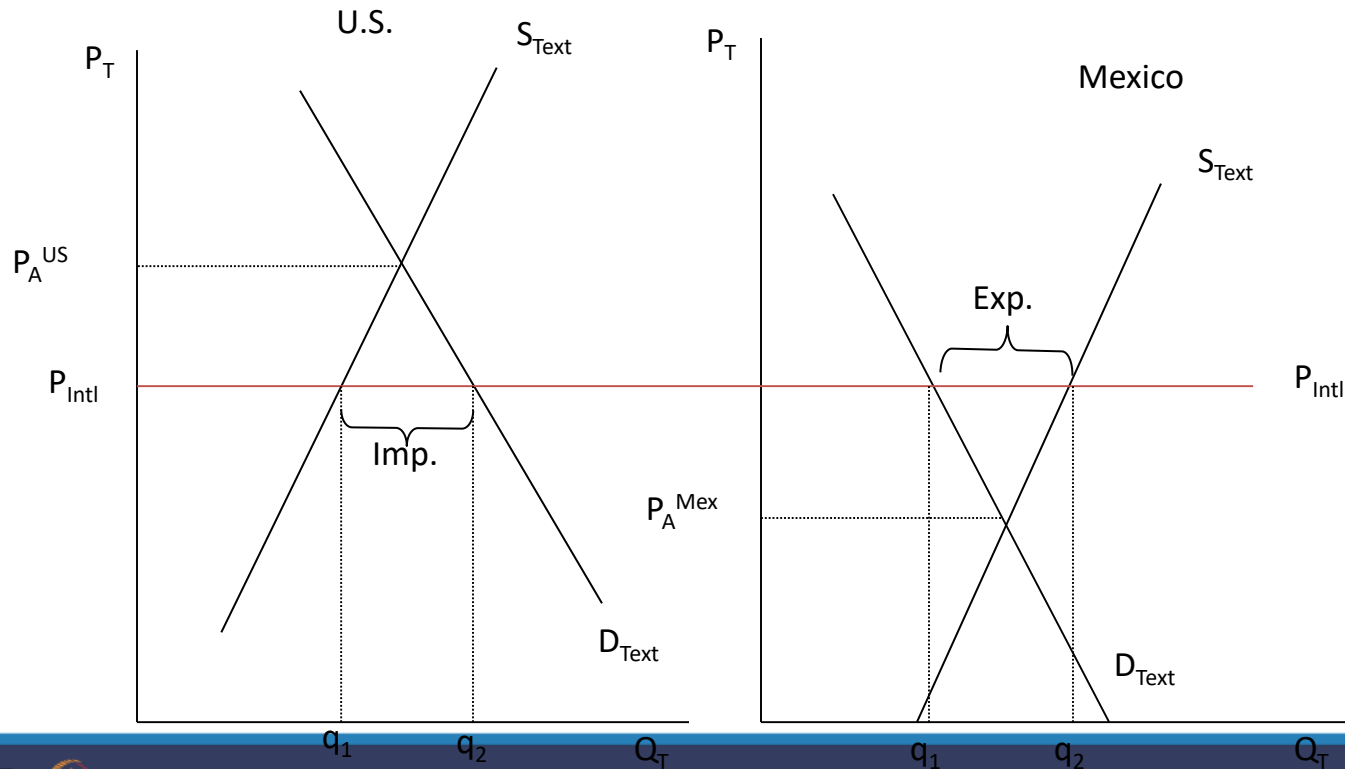


Qualification #3: Transportation Costs

- In the real world, it is costly to transport goods internationally
- How do the implications of our model change if we allow for transportation costs?
- Consider the supply and demand curves for textiles in Mexico and the U.S.:



Basic Set-Up



Adding Transportation Costs

- Suppose Mexico tries to pass along 100% of the transportation costs to the U.S. consumers
- In this case, the U.S. textile price will rise, and the quantity of imports demanded will fall
- Mexico will have a surplus, and will eventually lower their domestic price

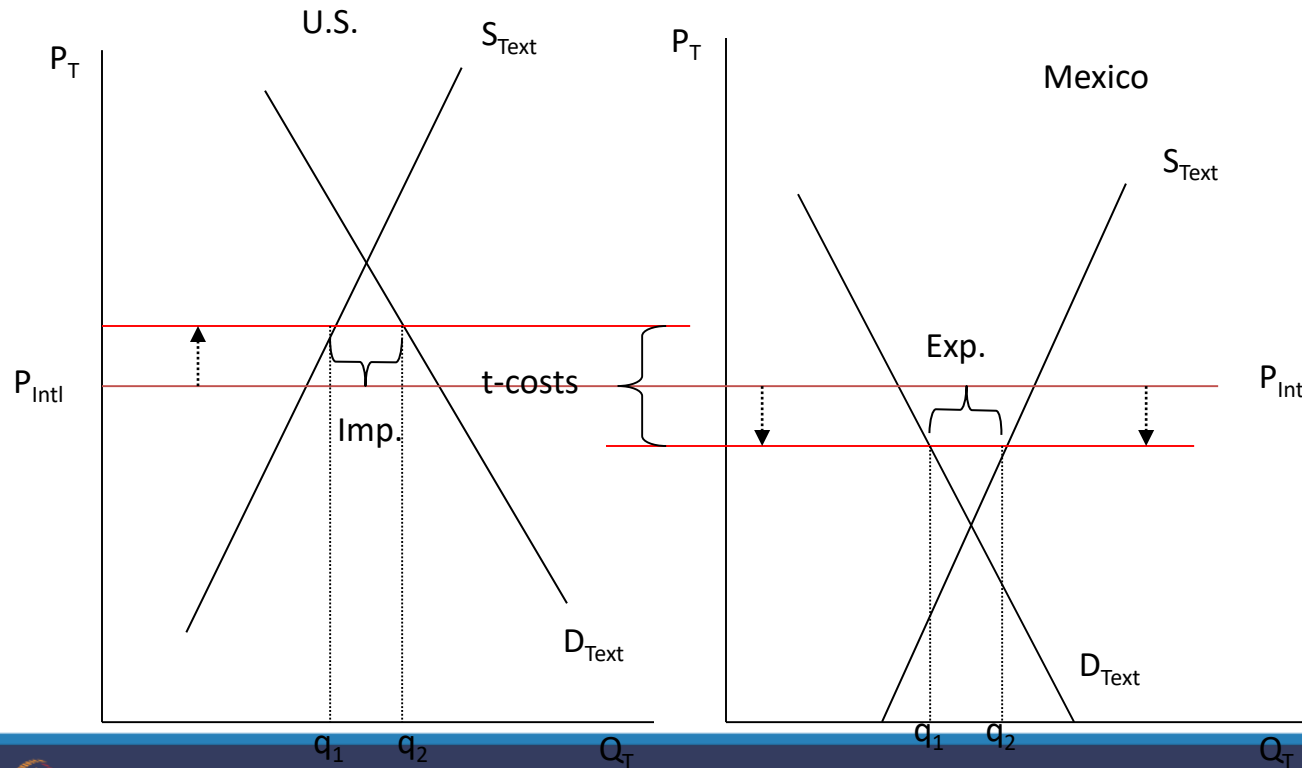


Adding Transportation Costs

- Therefore, unless Mexico is the only seller in the world, transportation costs will be borne by both the consumer (the U.S.) and the seller (Mexico).
- How does this look on the graph?



Adding Transportation Costs



Adding Transportation Costs

- Notice: there is no longer a single international textile price, but rather two different ones
- That is, transportation costs drive a wedge between the two countries' prices



Adding Transportation Costs: the Bottom Line

- In general, the H-O theorem will still hold
- The FPE theorem breaks down, since factor prices only equalize if the commodity prices do
- Therefore, in the presence of transportation costs, factor prices have a tendency to move towards each other, but we should expect equalization



Relaxing Other Assumptions

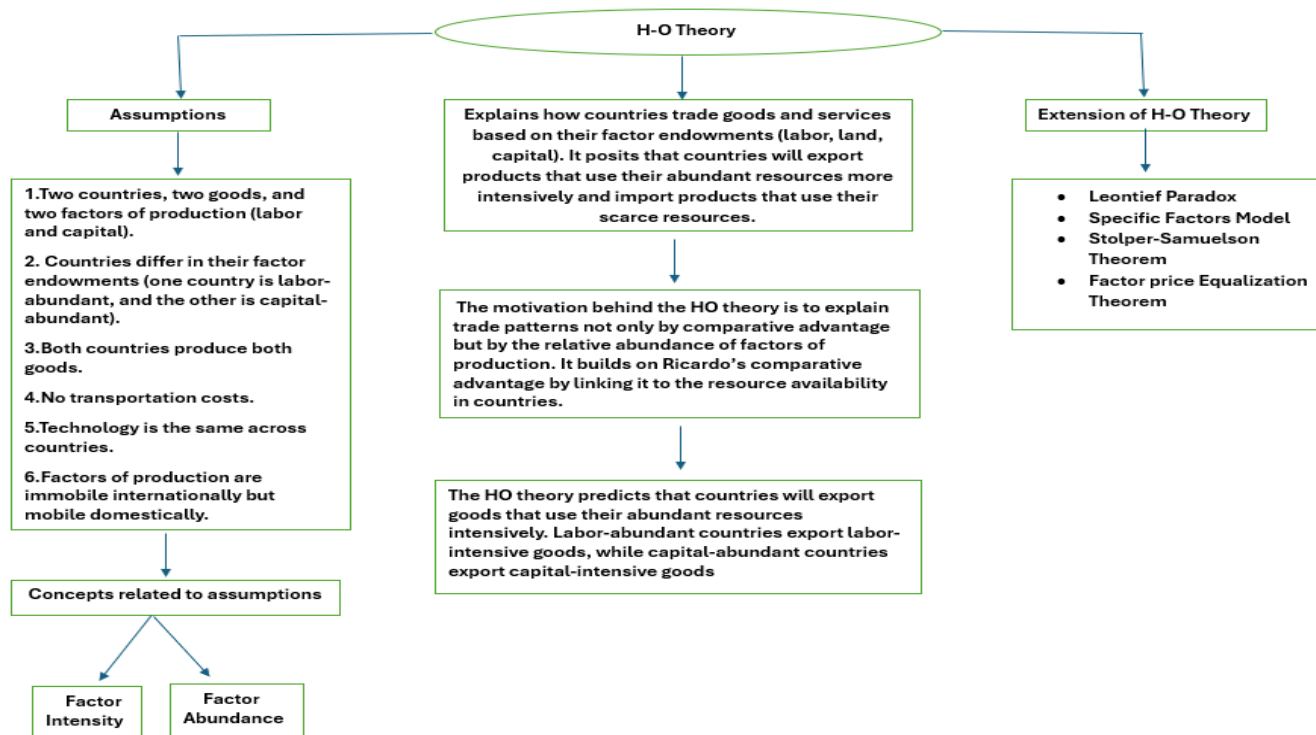
- One can relax many other assumptions and examine how the implications of the model change:
 - perfect competition
 - CRTS
 - identical production technologies
 - lack of policy obstacles
 - factors being perfectly transferable



Relaxing Other Assumptions

- These issues have involved many researchers for many years
- You can read a bit about this research in your text, but we have other fish to fry
- Suffice it to say that relaxing these assumptions can modify the basic H-O-S model, but doesn't lead us to scrap the model





*Thank
you*

